# Nonparametric ANOVA Notes: One-Way and Two-Way Layouts (Kruskal–Wallis, Friedman, Skillings–Mack, Multiple Comparisons, Median Polish)

## 1 Overview: What is "ANOVA" in Nonparametric Settings?

### 1.1 Goal

In ANOVA-style problems, we compare *three or more* populations. The classical (parametric) ANOVA focuses on comparing means under assumptions such as normality, equal variances, and independence.

More precisely, under the classical one-way ANOVA model,

$$X_{ij} = \mu_j + \varepsilon_{ij}, \qquad \varepsilon_{ij} \sim \text{i.i.d. } N(0, \mathbf{\Sigma}^2),$$

the primary goal is to test

$$H_0 : \mu_1 = \cdots = \mu_k,$$

by comparing between-group and within-group variability.

In contrast, the nonparametric ANOVA framework emphasizes

- weaker distributional assumptions,
- robustness to outliers and heavy-tailed distributions,
- invariance under monotone transformations,
- inference based on ranks rather than raw measurements.

A key advantage is that the procedures remain valid under a broad class of distributions, provided certain mild conditions such as continuity and independence hold.

Typically, the primary target is a *location parameter*, often interpreted as a median (or more generally a shift parameter). Rather than assuming

$$X_{ij} \sim N(\mu_j, \sigma^2),$$

we consider a more flexible model such as

$$X_{ij} = \theta + \tau_j + e_{ij},$$

where $e_{ij}$ are i.i.d. from a continuous distribution with median 0, but otherwise unspecified.

**Interpretation of the model.**

- $\theta$ represents a common baseline location (overall median),
- $\tau_j$ represents the treatment-specific deviation from the baseline,
- $e_{ij}$ captures random variation with no parametric restriction.

This formulation is often referred to as a *location-shift model.* Under $H_0$, all $\tau_j$ are equal, implying that all groups share the same distribution.

Equivalently, one may express the model in terms of distribution functions:

$$F_j(x) = F(x - \tau_j),$$

for some unknown continuous distribution function $F$. Thus, all group distributions differ only by a horizontal shift.

Thus, nonparametric ANOVA addresses questions like:

> Do the populations differ in location, without assuming a specific parametric form?

Importantly, this question can be interpreted in multiple equivalent ways:

- equality of medians,
- equality of distribution functions,
- absence of stochastic dominance among groups.

The most common tool in this setting is the Kruskal–Wallis test (for general differences), with related procedures for ordered or structured alternatives.

These methods are typically *rank-based* and therefore:

- distribution-free under the null hypothesis,
- robust to scale differences (to some extent),
- asymptotically efficient under a wide class of alternatives.

## 1.2 Variance Decomposition Viewpoint

Classical ANOVA is motivated by decomposing total variability:

$$\text{Total SS} = \text{Between-group SS} + \text{Within-group SS}.$$

This decomposition arises from the identity

$$\sum_{i,j}(X_{ij} - \bar{X})^2 = \sum_j n_j(\bar{X}_j - \bar{X})^2 + \sum_{i,j}(X_{ij} - \bar{X}_j)^2.$$

**Interpretation.**

- The *between-group sum of squares* measures systematic differences across groups (treatment effect).
- The *within-group sum of squares* measures random variability (noise).

The classical ANOVA $F$-statistic compares these two sources of variation.

In nonparametric ANOVA, the same structure remains, but it is applied to *ranks* instead of raw observations.

Let $R_{ij}$ denote the rank of $X_{ij}$ in the pooled sample of size $N$. Then:

- ranks replace observations,
- average ranks replace sample means,
- variability of ranks replaces variability of data.

Define the average rank in group $j$ as

$$\bar{R}_j = \frac{1}{n_j} \sum_{i=1}^{n_j} R_{ij}.$$

Under the null hypothesis (all distributions identical),

$$\mathbb{E}(\bar{R}_j) = \frac{N+1}{2},$$

since ranks are uniformly distributed over $\{1, \ldots, N\}$.

The Kruskal-Wallis statistic can be interpreted as

$$\text{Between-group variation of ranks} \quad \text{relative to} \quad \text{Total rank variation.}$$

More explicitly, one can write

$$H \propto \sum_{j=1}^{k} n_j (\bar{R}_j - \bar{R})^2,$$

where $\bar{R} = (N+1)/2$ is the grand mean rank.

Thus, the analogy with classical ANOVA is:

| | Classical ANOVA | Nonparametric ANOVA |
|---|---|---|
| Data | $X_{ij}$ | $R_{ij}$ |
| Group summary | $\bar{X}_j$ | $\bar{R}_j$ |
| Grand mean | $\bar{X}$ | $(N+1)/2$ |
| Variation | squared deviations | squared rank deviations |
| Test | $F$-statistic | Kruskal–Wallis $H$ |

Thus:

- Classical ANOVA partitions squared deviations of data.
- Nonparametric ANOVA partitions squared deviations of ranks.

A key advantage of this transformation is that it removes dependence on the scale and distribution of the data, while preserving ordering information.

The experimental design structure (independence, grouping, blocking) remains essential. What changes is the metric used to measure variation.

This perspective highlights an important principle:

> Nonparametric ANOVA preserves the design logic of ANOVA, while replacing parametric distributional assumptions with rank-based inference.

From a broader perspective, rank-based methods can be viewed as

- permutation-based procedures,
- invariant under strictly increasing transformations,
- approximating likelihood-based inference under semiparametric models.

This provides both robustness and theoretical justification for their widespread use.

# 2 One-Way Layout: $k$ Independent Samples (Kruskal–Wallis Framework)

## 2.1 Data Structure

We observe $k$ independent samples:

$$X_{1j}, \dots, X_{n_j j}, \qquad j = 1, \dots, k,$$

with total sample size

$$N = \sum_{j=1}^{k} n_j.$$

A standard one-way location-shift model is

$$X_{ij} = \theta + \tau_j + e_{ij}, \qquad i = 1, \dots, n_j,\ j = 1, \dots, k,$$

where

- $\theta$ is an overall location parameter (often interpreted as a common median),
- $\tau_j$ is the treatment effect for group $j$,
- $e_{ij}$ are i.i.d. random errors from a continuous distribution with median 0.

**Interpretation.** The model assumes that groups differ only by a location shift. No parametric form (e.g., normality) is imposed on the error distribution. Continuity ensures no ties in the population model.

Equivalently, one may write the model in terms of distribution functions:

$$F_j(x) = F(x - \tau_j),$$

for some unknown continuous distribution $F$. Thus, all group distributions differ only through a shift parameter.

Under $H_0$, all $\tau_j$ are equal, implying

$$F_1 = \cdots = F_k,$$

and hence the pooled sample is exchangeable.

### 2.2 Hypotheses (General Alternative)

$$H_0: \ \tau_1 = \tau_2 = \cdots = \tau_k \qquad \text{vs} \qquad H_A: \ \text{not all } \tau_j \text{ are equal.}$$

Under $H_0$, all $k$ groups share the same distribution. Hence the $N$ pooled observations are exchangeable, and any rank differences are due to chance. The alternative is non-directional: it does not specify which groups differ.

From a stochastic ordering perspective, $H_0$ implies that for any two groups $j$ and $\ell$,

$$\Pr(X_{ij} \le X_{i'\ell}) = \frac{1}{2}.$$

### 2.3 Kruskal-Wallis Test (General Alternative)

Pool all $N$ observations and assign ranks:

$$r_{ij} = \text{rank of } X_{ij} \text{ in the pooled sample.}$$

When ties occur, midranks are used.

Define the rank sum and average rank for group $j$:

$$R_j = \sum_{i=1}^{n_j} r_{ij}, \qquad \bar{R}_j = \frac{R_j}{n_j}.$$

The Kruskal–Wallis statistic is

$$H = \frac{12}{N(N+1)} \sum_{j=1}^{k} n_j \left( \bar{R}_j - \frac{N+1}{2} \right)^2. \tag{1}$$

**Equivalent form.** Using $R_j = n_j \bar{R}_j$, we may write

$$H = \frac{12}{N(N+1)} \sum_{j=1}^{k} \frac{R_j^2}{n_j} - 3(N+1).$$

This form is often useful for computation and theoretical derivations.

**Intuition.** Under $H_0$,

$$\mathbb{E}(\bar{R}_j) = \frac{N+1}{2}.$$

If some groups contain systematically larger (or smaller) observations, their average ranks shift away from $(N+1)/2$, inflating $H$. Thus $H$ measures the between-group variability of ranks, analogous to ANOVA but based on ranks rather than raw data.

**Moment calculations under $H_0$.** Under the null hypothesis:

$$\mathbb{E}(R_j) = n_j \frac{N+1}{2},$$

$$\mathrm{Var}(R_j) = \frac{n_j(N-n_j)(N+1)}{12}.$$

Moreover, for $j \neq \ell$,

$$\mathrm{Cov}(R_j, R_\ell) = -\frac{n_j n_\ell (N+1)}{12}.$$

These covariance relations imply that the vector $(R_1, \dots, R_k)$ lies in a $(k-1)$-dimensional subspace, explaining the degrees of freedom in the asymptotic distribution.

**Quadratic form representation.** Let $\mathbf{R} = (R_1, \dots, R_k)^T$. Then $H$ can be written as a centered quadratic form:

$$H \propto (\mathbf{R} - \mathbb{E}[\mathbf{R}])^T \mathbf{\Sigma}^{-1} (\mathbf{R} - \mathbb{E}[\mathbf{R}]),$$

where $\mathbf{\Sigma}$ is the covariance matrix of rank sums under $H_0$. This representation makes the asymptotic $\chi^2$ distribution transparent.

### 2.4 Reference Distribution

- **Exact (Permutation).** Under $H_0$, all assignments of ranks to groups are equally likely. The null distribution of $H$ may therefore be obtained by permuting group labels.

  This relies on the exchangeability of observations under $H_0$.

- **Large-Sample Approximation.** For moderate sample sizes and no severe ties,

  $$H \xrightarrow{d} \chi^2_{k-1}.$$

  This follows from a multivariate central limit theorem applied to the rank sums, together with the quadratic form representation above.

  The degrees of freedom $k-1$ arise because the $k$ group mean ranks satisfy one linear constraint:

  $$\sum_{j=1}^{k} n_j \bar{R}_j = \sum_{i,j} r_{ij} = \frac{N(N+1)}{2}.$$

## 2.5 Ties

If ties are present, the variance of ranks is reduced. A common correction factor is

$$C = 1 - \frac{\sum_m (t_m^3 - t_m)}{N^3 - N},$$

where $t_m$ is the size of the $m$th tie group. The corrected statistic is

$$H^* = \frac{H}{C}.$$

Permutation methods remain valid under ties.

## 2.6 Example: Mucociliary Clearance Half-Times

Three groups: normal, obstructive airway disease, asbestosis.

```
normal     = c(2.9, 3.0, 2.5, 2.6, 3.2)
obstruct   = c(3.8, 2.7, 4.0, 2.4)
asbestosis = c(2.8, 3.4, 3.7, 2.2, 2.0)

x = c(normal, obstruct, asbestosis)

treatment = factor(c(rep("normal", length(normal)),
                     rep("obstruct", length(obstruct)),
                     rep("asbestosis", length(asbestosis))))

KW = kruskal.test(x ~ treatment)
print(KW)
```

For these data,

$$H \approx 0.7714, \qquad p \approx 0.6800,$$

so we fail to reject $H_0$ at common significance levels.

**Interpretation.** There is no statistical evidence that the three groups differ in location. In particular, the observed rank differences are consistent with random variation under exchangeability.

## 2.7 Other Structured Alternatives in One-Way Layouts

### Ordered Alternatives (Jonckheere–Terpstra)

If treatments have a natural order (e.g., increasing dose), we test

$$H_A: \ \tau_1 \le \tau_2 \le \cdots \le \tau_k, \quad \text{with at least one strict inequality.}$$

The test aggregates Mann–Whitney "win" counts across all $\binom{k}{2}$ ordered treatment pairs. It has greater power when a monotone trend exists.

### Umbrella Alternatives (Mack–Wolfe)

If the response increases up to a peak at $p$ and then decreases,

$$H_A: \ \tau_1 \leq \cdots \leq \tau_{p-1} \leq \tau_p \geq \tau_{p+1} \geq \cdots \geq \tau_k,$$

with at least one strict inequality. The test emphasizes pairwise comparisons consistent with the umbrella structure.

### Treatments Versus a Control (Fligner–Wolfe)

Let treatment 1 denote the control. We test

$$H_0: \ \tau_i = \tau_1, \ i = 2, \ldots, k \quad \text{vs} \quad H_A: \ \tau_i \geq \tau_1, \ i = 2, \ldots, k,$$

with at least one strict inequality.

All observations are ranked, and the test statistic is based on the sum of ranks among the noncontrol treatments. This approach is particularly useful in dose-control studies.

```
# install.packages(c("PMCMRplus", "DescTools"))

library(PMCMRplus)

library(DescTools)

# x = response, g = ordered treatment factor
g <- factor(treatment, levels = c("normal", "obstruct", "asbestosis"),
            ordered = TRUE)

jonckheereTest(x, g, alternative = "greater")

JonckheereTerpstraTest(x, g, alternative = "increasing")

# install.packages("PMCMRplus")
library(PMCMRplus)

############################################################
# 3. Umbrella Alternative (Mack-Wolfe)
############################################################
# Order the groups and specify the peak group
g <- factor(treatment,
            levels = c("normal", "obstruct", "asbestosis"),
            ordered = TRUE)

# Example: peak at the 2nd group ("obstruct")
mackWolfeTest(x, g, p = 2, nperm = 10000)

# install.packages("NSM3")
library(NSM3)
```

```
# known peak at group 2
pUmbrPK(x, 2, g, method = "Asymptotic")

# unknown peak
pUmbrPU(x, g, method = "Monte Carlo", n.mc = 10000)

# install.packages("PMCMRplus")
library(PMCMRplus)

# put control first in the factor order
g <- factor(treatment,
            levels = c("normal", "obstruct", "asbestosis"))

FW = flignerWolfeTest(x, g, alternative = "greater")
print(FW)
```

# 3 Multiple Comparisons After Kruskal–Wallis

## 3.1 Why Multiple Comparisons?

The Kruskal-Wallis (KW) test addresses the global null hypothesis

$$H_0: \ \tau_1 = \tau_2 = \cdots = \tau_k.$$

If $H_0$ is rejected, we conclude that *at least one* group differs, but the test does not identify *which* groups are different.

In practice, scientific questions are often pairwise:

- Which treatments differ from each other?
- Is each treatment different from control?
- Are groups naturally clustered?

Thus, after rejecting the global null, we perform *multiple comparisons.* However, testing many hypotheses inflates the probability of false positives unless properly controlled.

Formally, if multiple hypotheses are tested simultaneously, the probability of incorrectly rejecting at least one true null hypothesis increases with the number of tests. This motivates the need for error rate control.

## 3.2 Familywise Error Rate (FWER)

Suppose we test $s$ null hypotheses

$$H_0^{(1)}, \ldots, H_0^{(s)}.$$

Define:

$$\alpha_I = \text{per-comparison Type I error rate}, \qquad \alpha_E = \text{experiment-wise (familywise) error rate.}$$

The familywise error rate (FWER) is

$$\alpha_E = \Pr\left(\bigcup_{i=1}^{s}\{\text{reject } H_0^{(i)} \text{ when it is true}\}\right).$$

Equivalently,

$$\alpha_E = \Pr(V \geq 1),$$

where $V$ is the number of false rejections.

If the $s$ tests were independent,

$$\alpha_E = 1 - (1 - \alpha_I)^s.$$

Solving for $\alpha_I$ gives

$$\alpha_I = 1 - (1 - \alpha_E)^{1/s}.$$

When $\alpha_E$ is small and $s$ is moderate,

$$\alpha_I \approx \frac{\alpha_E}{s}.$$

Importantly, the independence assumption rarely holds in practice, especially for rank-based tests where statistics are computed from the same dataset. Thus, valid procedures must control FWER under arbitrary dependence.

**Bonferroni Correction.** A simple and always valid (possibly conservative) rule:

$$\alpha_I \leq \frac{\alpha_E}{s}.$$

Equivalently, reject $H_0^{(i)}$ if

$$p_i \leq \frac{\alpha_E}{s}.$$

Bonferroni follows from the union bound:

$$\Pr\left(\bigcup_{i=1}^{s} A_i\right) \leq \sum_{i=1}^{s} \Pr(A_i),$$

and therefore controls FWER under arbitrary dependence.

**Holm's Step-Down Procedure.** Holm (1979) improves upon Bonferroni while still controlling FWER.

1. Order the $p$-values:
$$p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(s)}.$$
2. Compare sequentially:
$$p_{(i)} \leq \frac{\alpha_E}{s - i + 1}.$$
3. Stop at the first non-rejection.

**Interpretation.**

- The smallest $p$-value is tested most stringently.
- Subsequent tests become less stringent as hypotheses are rejected.
- This reflects a *step-down* control of error.

Holm is uniformly more powerful than Bonferroni while maintaining strong FWER control.

**Connection to closure principle.** Holm's procedure can be viewed as a shortcut to the closure principle, which tests all intersection hypotheses and guarantees strong control of FWER.

### 3.3 Pairwise Testing via Wilcoxon Rank-Sum

In a one-way layout with $k$ groups, the number of pairwise comparisons is

$$s = \binom{k}{2} = \frac{k(k-1)}{2}.$$

After rejecting KW:

1. For each pair $(i, j)$, conduct a two-sample Wilcoxon rank-sum (Mann–Whitney) test.
2. Obtain the $p$-value $p_{ij}$.
3. Adjust $\{p_{ij}\}$ using Bonferroni, Holm, or another FWER-controlling method.

**Rank-based interpretation.** Each Wilcoxon test is based on pairwise comparisons:

$$U_{ij} = \sum \mathbf{1}(X_{i\ell} < X_{jm}),$$

which estimates

$$\Pr(X_i < X_j).$$

Thus, pairwise comparisons after KW correspond to identifying which group pairs exhibit stochastic dominance.

This approach is intuitive because:

- KW tests a global equality hypothesis.
- Wilcoxon tests assess specific pairwise shifts.
- Both procedures are rank-based and distribution-free under continuity.

### 3.4 Contrast Language

In linear-model language, a *contrast* is a linear combination

$$C = a_1\tau_1 + \cdots + a_k\tau_k, \qquad \sum_{j=1}^{k} a_j = 0.$$

Pairwise differences correspond to contrasts such as

$$C = \tau_i - \tau_j.$$

There are

$$\binom{k}{2}$$

distinct pairwise contrasts.

Thus, multiple comparisons after KW can be interpreted as testing all pairwise contrasts under rank-based inference.

From a semiparametric perspective, these tests estimate differences in location parameters without specifying the underlying distribution.

### 3.5 Example: Gizzard Shad Lengths

Suppose $k = 4$ groups and experiment-wise level

$$\alpha_E = 0.01.$$

Then

$$s = \binom{4}{2} = 6.$$

```
num.of.contrasts = 4*(4-1)/2

library(NSM3)
data(gizzards)

grp = factor(c(rep("I", length(gizzards[[1]])),
               rep("II", length(gizzards[[2]])),
               rep("III", length(gizzards[[3]])),
               rep("IV", length(gizzards[[4]]))))

leng = as.numeric(unlist(gizzards))

kw.test = kruskal.test(leng, grp)
kw.test$p.value

p.value12 = wilcox.test(gizzards[[1]], gizzards[[2]])$p.value
p.value13 = wilcox.test(gizzards[[1]], gizzards[[3]])$p.value
p.value14 = wilcox.test(gizzards[[1]], gizzards[[4]])$p.value
p.value23 = wilcox.test(gizzards[[2]], gizzards[[3]])$p.value
```

```
p.value24 = wilcox.test(gizzards[[2]], gizzards[[4]])$p.value
p.value34 = wilcox.test(gizzards[[3]], gizzards[[4]])$p.value

round(c(p.value12,p.value13,p.value14,
        p.value23,p.value24,p.value34), 3)

round(p.adjust(c(p.value12,p.value13,p.value14,
                 p.value23,p.value24,p.value34),
               method="bonferroni"), 3)

round(p.adjust(c(p.value12,p.value13,p.value14,
                 p.value23,p.value24,p.value34),
               method="holm"), 3)
```

Interpretation (at experiment-wise level 0.01):

- The global KW test rejects equality.
- Bonferroni-adjusted $p$-values indicate that groups I and II are similar, groups III and IV are similar, but the first pair differs from the second pair.
- Holm adjustment leads to the same conclusion in this example.

Hence the data support a grouping structure of the form

$$(\tau_1 = \tau_2) \neq (\tau_3 = \tau_4).$$

### 3.6 Summary

- Kruskal–Wallis answers: *Is there any difference?*
- Multiple comparisons answer: *Where are the differences?*
- FWER control ensures the probability of at least one false discovery remains bounded by $\alpha_E$.
- Bonferroni is simple and conservative; Holm improves power while maintaining validity.

## 4 Two-Way Layout: Treatment + Blocking Factor (Randomized Block Designs)

### 4.1 Data Structure and Model

In a two-way layout with blocking, we study two factors:

- **Treatment factor** $A$ with $k$ levels (primary factor of interest).
- **Blocking factor** $B$ with $n$ levels (nuisance factor used to reduce variability).

In a *randomized complete block design* (RCBD):

- Each block contains all $k$ treatments exactly once.

- There is one observation per treatment per block.
- Total sample size is $N = nk$.

The nonparametric additive location model is

$$X_{ij} = \theta + \beta_i + \tau_j + e_{ij}, \qquad i = 1, \ldots, n, \ j = 1, \ldots, k,$$

where:

- $\theta$ is an overall location parameter,
- $\beta_i$ is the block effect (captures systematic differences across blocks),
- $\tau_j$ is the treatment effect of interest,
- $e_{ij}$ are i.i.d. errors from a continuous distribution with median 0.

**Interpretation.** Blocking removes between-block variability from the error term, increasing sensitivity to treatment differences. The goal is inference on $\tau_j$ while adjusting for $\beta_i$.

Equivalently, one may write

$$F_{ij}(x) = F(x - \beta_i - \tau_j),$$

so that each observation is a shifted version of a common distribution. The block effect $\beta_i$ acts as a nuisance location shift shared across treatments within the same block.

This setup is closely related to repeated-measures designs, where each block corresponds to a subject and treatments correspond to repeated conditions.

The null hypothesis is

$$H_0 : \ \tau_1 = \tau_2 = \cdots = \tau_k.$$

## 4.2 RCBD and the Friedman Test

The Friedman test is the nonparametric analogue of two-way ANOVA with blocks (no interaction term).

**Rank Within Blocks.** For each block $i$, rank the $k$ observations

$$X_{i1}, \ldots, X_{ik}$$

from smallest to largest.

Let $r_{ij}$ denote the rank of treatment $j$ in block $i$. If ties occur within a block, use midranks.

**Key point:** Ranking is performed *within each block*. This removes the block effect $\beta_i$ because ranks are invariant under additive shifts.

Formally, for any constant $c$,

$$\text{rank}(X_{ij}) = \text{rank}(X_{ij} + c),$$

so the term $\beta_i$ is eliminated in the ranking step.

**Sum Ranks Across Blocks.** Define

$$R_j = \sum_{i=1}^{n} r_{ij}, \qquad \bar{R}_j = \frac{R_j}{n}.$$

Under $H_0$, each treatment should have average rank

$$\mathbb{E}(\bar{R}_j) = \frac{k+1}{2}.$$

**Moment structure under $H_0$.** Within each block,

$$\mathbb{E}(r_{ij}) = \frac{k+1}{2}, \qquad \mathrm{Var}(r_{ij}) = \frac{k^2-1}{12}.$$

Summing over blocks:

$$\mathbb{E}(R_j) = n\frac{k+1}{2}, \qquad \mathrm{Var}(R_j) = n\frac{k^2-1}{12}.$$

Moreover, within each block,

$$\sum_{j=1}^{k} r_{ij} = \frac{k(k+1)}{2},$$

which induces dependence across treatments.

**Test Statistic.** The Friedman statistic is

$$S = \frac{12n}{k(k+1)} \sum_{j=1}^{k} \left(\bar{R}_j - \frac{k+1}{2}\right)^2. \tag{2}$$

**Equivalent form.** Using $R_j = n\bar{R}_j$, we can write

$$S = \frac{12}{nk(k+1)} \sum_{j=1}^{k} R_j^2 - 3n(k+1).$$

**Interpretation.**

- If treatments are equivalent, average ranks fluctuate randomly.
- If a treatment tends to be larger within blocks, its average rank will be systematically higher.
- $S$ measures between-treatment variation of ranks, after removing block effects.

**Quadratic form representation.** Let $\mathbf{R} = (R_1, \ldots, R_k)^T$. Then

$$S \propto (\mathbf{R} - \mathbb{E}[\mathbf{R}])^T \boldsymbol{\Sigma}^{-1} (\mathbf{R} - \mathbb{E}[\mathbf{R}]),$$

where $\boldsymbol{\Sigma}$ is the covariance matrix induced by within-block ranking. This parallels the Kruskal–Wallis formulation.

**Reference Distribution.**

- **Exact (Permutation).** Within each block, all $k!$ permutations of ranks are equally likely under $H_0$. Thus inference is based on permutations within blocks.
- **Large-Sample Approximation.**
  For large $n$ (number of blocks),
  $$S \,\dot{\sim}\, \chi^2_{k-1}.$$
  This follows from a multivariate central limit theorem applied to rank sums across blocks.

Tie corrections are available if needed.

## 4.3 Connection with Variance Decomposition

Classical two-way ANOVA decomposes variability into:

$$\text{Total} = \text{Treatment} + \text{Block} + \text{Error}.$$

The Friedman test follows the same design logic, but operates on ranks within blocks:

- Ranking within block removes $\beta_i$ automatically.
- Variation of rank sums across treatments measures treatment effect.

Equivalently, one can view Friedman as performing ANOVA on transformed data:

$$X_{ij} \mapsto r_{ij}.$$

Thus Friedman is the rank-based analogue of two-way ANOVA without interaction.

## 4.4 Example: Rounding First Base

Blocks = subjects; Treatments = rounding methods.

```
library(magrittr)
library(NSM3)
library(dplyr)
library(tidyr)

data("rounding.times")

df = rounding.times %>% data.frame
colnames(df) = paste0("Method_", seq(1,3))
df = mutate(df, block = seq(1, dim(df)[1]))
df = gather(df, key = "Method", value = "xij", c(1:3))

friedman.test(df$xij, df$Method, df$block)
```

Reported result:

$$\chi^2 \approx 11.14, \qquad p \approx 0.0038.$$

Conclusion: Reject $H_0$. There is evidence that at least one rounding method differs.

**Interpretation.** The observed differences in within-block ranks are too large to be explained by chance alone, indicating systematic treatment effects.

### 4.5 Multiple Comparisons in RCBD

After rejecting Friedman, we identify which treatments differ.

Because observations are paired within blocks, comparisons are based on within-block differences.

For each pair $(i, j)$:

$$D_b = X_{bi} - X_{bj}, \qquad b = 1, \ldots, n.$$

We may apply:

- Sign test,
- Wilcoxon signed-rank test,

followed by FWER control (e.g., Holm adjustment).

These procedures exploit the paired structure of the data, analogous to paired $t$-tests in parametric settings.

```
library(BSDA)

p.value12 = SIGN.test(x = c(rounding.times[,1]-
                            rounding.times[,2]),
                      alternative = "two.sided")$p.value

p.value13 = SIGN.test(x = c(rounding.times[,1]-
                            rounding.times[,3]),
                      alternative = "two.sided")$p.value

p.value23 = SIGN.test(x = c(rounding.times[,2]-
                            rounding.times[,3]),
                      alternative = "two.sided")$p.value

round(c(p.value12, p.value13, p.value23), 3)
round(p.adjust(c(p.value12, p.value13, p.value23),
               method="holm"), 3)
```

Holm adjustment controls experiment-wise error rate.

### 4.6 Treatments Versus a Control in RCBD

Sometimes one treatment is a control (say $j = 1$). We test

$$H_0: \ \tau_j = \tau_1 \qquad \text{vs} \qquad H_A: \ \tau_j > \tau_1.$$

For each treatment $j > 1$, compute paired differences

$$D_b = X_{bj} - X_{b1}.$$

Apply one-sided sign or signed-rank tests, then adjust $p$-values via Holm.

This framework is especially useful when interest is in improvement relative to a baseline.

This approach is common in:

- Clinical crossover trials,
- Agricultural experiments,
- Behavioral or speech adaptation studies.

### 4.7 Summary

- Blocking reduces variability and increases power.
- Friedman is the rank-based analogue of two-way ANOVA.
- Ranking within blocks removes block effects automatically.
- Pairwise comparisons use paired nonparametric tests.
- Holm adjustment controls FWER.

## 5 Incomplete Block and General Two-Way Layouts: Skillings–Mack

### 5.1 When RCBD is Not Available

The Friedman test applies to a *randomized complete block design* (RCBD), where each block contains all $k$ treatments exactly once.

In practice, however, experiments are often *incomplete*:

- Some treatments are missing in certain blocks.
- The number of treatments per block may vary.
- Observations may be unbalanced across cells.

In such situations:

- Ranking within blocks is still meaningful.
- But the classical Friedman statistic is no longer valid, because its null distribution relies on equal treatment representation.

To address this, the **Skillings–Mack test** generalizes the Friedman test to incomplete block designs.

## 5.2 Model Framework

Suppose:

- $k$ treatments,
- $n$ blocks,
- Not all $X_{ij}$ are observed.

The additive nonparametric model remains

$$X_{ij} = \theta + \beta_i + \tau_j + e_{ij},$$

with the same interpretation as before:

- $\beta_i$ = block effect,
- $\tau_j$ = treatment effect of interest,
- $e_{ij}$ = i.i.d. errors with median 0.

We test

$$H_0 : \ \tau_1 = \cdots = \tau_k.$$

Let $\mathcal{J}_i \subset \{1, \ldots, k\}$ denote the set of treatments observed in block $i$, and let $k_i = |\mathcal{J}_i|$ denote the number of observed treatments in that block.

## 5.3 Idea of the Skillings–Mack Test

The Skillings–Mack procedure extends the Friedman logic:

1. Within each block $i$, rank only the *observed* treatments
$$\{X_{ij} : j \in \mathcal{J}_i\}.$$
2. Let $r_{ij}$ denote the rank of $X_{ij}$ among these $k_i$ observations.
3. Center ranks within each block:
$$r_{ij}^c = r_{ij} - \frac{k_i + 1}{2}.$$
4. Apply a normalization to account for varying block sizes:
$$a_{ij} = \frac{r_{ij}^c}{\sqrt{\frac{k_i^2 - 1}{12}}}.$$
5. Aggregate across blocks:
$$T_j = \sum_{i : j \in \mathcal{J}_i} a_{ij}.$$

Thus, it preserves:

- Blocking structure,
- Rank-based inference,
- Distribution-free validity under continuity.

**Interpretation.**

- Centering removes the block effect $\beta_i$.
- Scaling ensures comparability across blocks with different sizes.
- Aggregation accumulates evidence of treatment effects across blocks.

**Test statistic.** ([Chatfield and Mander(2009)] has more details.) The Skillings-Mack statistic is constructed as a quadratic form:

$$S_{SM} = \mathbf{T}^T \mathbf{\Sigma}^{-1} \mathbf{T},$$

where $\mathbf{T} = (T_1, \ldots, T_k)^T$ and $\mathbf{\Sigma}$ is the covariance matrix of $\mathbf{T}$ under $H_0$.

In practice, a simplified form is used:

$$S_{SM} = \sum_{j=1}^{k} \frac{T_j^2}{\mathrm{Var}(T_j)},$$

with appropriate covariance adjustment.

The resulting statistic has, asymptotically,

$$S_{SM} \dot{\sim} \chi^2_{k-1}.$$

When the design is complete, the Skillings-Mack statistic reduces to the Friedman statistic.

## 5.4 Moment Structure and Dependence

Under $H_0$:

- $\mathbb{E}(a_{ij}) = 0$,
- $\mathrm{Var}(a_{ij}) = 1$,

but within each block,

$$\sum_{j \in \mathcal{J}_i} a_{ij} = 0,$$

which induces dependence across treatments.

Thus the covariance matrix $\mathbf{\Sigma}$ is singular of rank $k-1$, leading to the $\chi^2_{k-1}$ limiting distribution.

## 5.5 Permutation Interpretation

Under $H_0$, within each block $i$, all permutations of observed treatments are equally likely.

Thus:

- inference is based on permutations within blocks,
- the test is distribution-free under continuity,
- exchangeability holds conditional on observed treatment sets $\mathcal{J}_i$.

### 5.6 Example: Niacin in Bran Flakes

The lecture illustrates an incomplete block example involving niacin content across treatments.

```
library(Skillings.Mack)
data("niacin")

df = matrix(niacin, nrow=3, byrow=TRUE) %>%
     t %>%
     data.matrix()

colnames(df) = paste0("trt", c(0,4,8))

Ski.Mack.test = Ski.Mack(df)
```

Reported result:

$$S_{SM} \approx 22.68, \qquad p \approx 0.01.$$

At $\alpha = 0.05$, we reject $H_0$ and conclude that at least one treatment differs in location.

**Interpretation.** The observed deviations in normalized ranks are too large to be explained by random variation under the null hypothesis of equal treatment effects.

### 5.7 Interpretation and Advantages

- Extends Friedman to incomplete block designs.
- Handles unequal numbers of treatments per block.
- Maintains rank-based robustness.
- Automatically adjusts for differing block sizes.
- Asymptotically chi-square with $k - 1$ degrees of freedom.

Thus, the Skillings-Mack test provides a practical nonparametric alternative to two-way ANOVA when data are unbalanced or partially missing.

### 5.8 Summary

- Complete block $\Rightarrow$ Friedman.
- Incomplete block $\Rightarrow$ Skillings–Mack.
- Both preserve blocking and rank-based inference.
- Both test equality of treatment effects.

# 6 Additional Methods for Randomized Block Designs

## 6.1 Quade Test: Weighted Rank-Based Procedure for RCBD

**Motivation.** The Friedman test treats all blocks equally. However, in many experiments, some blocks exhibit stronger separation among treatments than others. The *Quade test* improves efficiency by incorporating block-specific weights based on within-block variability.

**Model.** Consider a randomized complete block design (RCBD):

$$X_{ij} = \theta + \beta_i + \tau_j + e_{ij}, \qquad i = 1, \ldots, b, \quad j = 1, \ldots, k,$$

where $\beta_i$ is a block effect and $\tau_j$ is the treatment effect. We test

$$H_0 : \ \tau_1 = \cdots = \tau_k.$$

**Procedure.**

1. For each block $i$, compute the within-block range:
$$Q_i = \max_j X_{ij} - \min_j X_{ij}.$$
2. Rank the ranges $\{Q_i\}$ across blocks to obtain weights $w_i$.
3. Within each block, rank the observations:
$$r_{ij} = \text{rank of } X_{ij} \text{ within block } i.$$
4. Center the ranks:
$$r_{ij}^c = r_{ij} - \frac{k+1}{2}.$$
5. Form weighted scores:
$$a_{ij} = w_i \, r_{ij}^c.$$
6. Aggregate across blocks:
$$T_j = \sum_{i=1}^{b} a_{ij}.$$

**Test statistic.** The Quade statistic is based on the variability of $\{T_j\}$ across treatments, and is approximately $F$-distributed under $H_0$.

**Interpretation.**

- Blocks with larger within-block variation receive greater weight.
- This can improve power relative to Friedman when block informativeness varies.

**R implementation.**

```
y <- c(8, 10, 12,
       7, 11,  9,
       6,  8, 10,
       9, 10, 13)

treatment <- factor(rep(c("A","B","C"), times = 4))
block <- factor(rep(1:4, each = 3))

quade.test(y, treatment, block)
```

## 6.2 Aligned Rank Transform (ART): Nonparametric Factorial ANOVA

**Motivation.** Classical rank-based methods such as Friedman and Quade are limited to simple designs and do not easily accommodate multiple factors or interactions. The *aligned rank transform (ART)* provides a general framework for nonparametric ANOVA in factorial designs.

**Model.** Consider a factorial model (e.g., treatment and block):

$$X_{ij} = \theta + \beta_i + \tau_j + e_{ij},$$

or more generally,

$$X = \text{main effects} + \text{interactions} + \text{error}.$$

**Key idea.** ART proceeds in two main steps:

1. **Alignment:** Remove nuisance effects to isolate the effect of interest. For example, to test treatment effects, subtract estimated block effects.
2. **Ranking:** Rank the aligned responses across all observations.
3. **ANOVA:** Apply a standard ANOVA to the ranked aligned responses.

**Interpretation.**

- Alignment isolates the contribution of a specific factor.
- Ranking ensures robustness and distribution-free inference.
- The final ANOVA provides familiar $F$-tests for main effects and interactions.

**Advantages.**

- Handles multiple factors and interactions.
- Extends rank-based inference beyond Friedman-type designs.
- Retains interpretability of ANOVA output.

**R implementation.**

```
library(ARTool)

df <- data.frame(
  y = c(8, 10, 12,
        7, 11,  9,
        6,  8, 10,
        9, 10, 13),
  treatment = factor(rep(c("A", "B", "C"), times = 4)),
  block = factor(rep(1:4, each = 3))
)

m <- art(y ~ treatment + (1|block), data = df)
anova(m)
```

**Example (factorial design).**

```
# install.packages("ARTool")
library(ARTool)

# Create a proper data.frame (required for ART)
df2 <- data.frame(
  y = c(8, 10, 12,
        7, 11,  9,
        6,  8, 10,
        9, 10, 13),
  A = factor(rep(c("A1", "A2"), each = 6)),
  B = factor(rep(c("B1", "B2", "B3"), times = 4))
)

# ART model with full factorial structure
m2 <- art(y ~ A * B, data = df2)

# ANOVA on aligned ranks
anova(m2)
```

### 6.3 Summary

- The **Quade test** extends Friedman by weighting blocks according to their informativeness, improving power when block variability differs.
- The **ART method** provides a general nonparametric ANOVA framework for factorial designs, allowing testing of main effects and interactions.
- In practice:
  - Use Friedman or Quade for simple RCBD settings,
  - Use ART when multiple factors or interactions are present.

# 7 Median Polish (Tukey): Robust Additive Fits for Two-Way Tables

## 7.1 Motivation

For a two-way table with rows $i = 1, \ldots, r$ and columns $j = 1, \ldots, c$, median polish provides a robust decomposition under an additive model:

$$X_{ij} = \theta + \beta_i + \tau_j + e_{ij},$$

where

- $\theta$ is an overall location (median) effect,
- $\beta_i$ is the row effect (e.g., region),
- $\tau_j$ is the column effect (e.g., education level),
- $e_{ij}$ are residuals.

Unlike classical two-way ANOVA (which uses means and sums of squares), median polish uses medians and is therefore:

- Robust to outliers,
- Less sensitive to skewness,
- Invariant to monotone transformations of scale (to some extent),
- Useful for exploratory data analysis (EDA),
- Appropriate when additivity is plausible but normality is questionable.

From a statistical perspective, median polish can be viewed as a robust alternative to least-squares estimation. Instead of minimizing

$$\sum_{i,j} e_{ij}^2,$$

it implicitly targets an $L_1$-type criterion based on medians, which is less sensitive to extreme observations.

The goal is not formal hypothesis testing, but rather a *robust decomposition* of systematic structure.

## 7.2 Algorithm: Iterative Median Subtraction

Initialize:

$$\beta_i^{(0)} = 0, \quad \tau_j^{(0)} = 0.$$

**Step 1: Overall Median** Compute the overall median:

$$\theta = \mathrm{median}_{i,j}(X_{ij}).$$

Subtract $\theta$ from every cell.

**Step 2: Row Effects** For each row $i$:

1. Compute row median:
$$m_i = \text{median}_j(X_{ij}^{\text{current}}).$$
2. Update row effect:
$$\beta_i \leftarrow \beta_i + m_i.$$
3. Subtract $m_i$ from every element in row $i$.

**Step 3: Column Effects** For each column $j$:

1. Compute column median:
$$m_j = \text{median}_i(X_{ij}^{\text{current}}).$$
2. Update column effect:
$$\tau_j \leftarrow \tau_j + m_j.$$
3. Subtract $m_j$ from every element in column $j$.

**Step 4: Iterate** Repeat row and column adjustments until:

$$\text{All row and column medians} \approx 0.$$

The remaining values are residuals $e_{ij}$.

Final decomposition:

$$X_{ij} = \theta + \beta_i + \tau_j + e_{ij}.$$

**Remarks on convergence.**

- The algorithm typically converges in a small number of iterations.
- Convergence corresponds to reaching a fixed point where row and column medians are zero.
- The solution is not necessarily unique, but is stable in practice.

### 7.3 Interpretation of Components

- $\theta$ = overall central tendency of the table.
- $\beta_i$ = how row $i$ deviates from overall median.
- $\tau_j$ = how column $j$ deviates from overall median.
- $e_{ij}$ = interaction or local deviation not explained by additivity.

The residuals $e_{ij}$ may be interpreted as an *interaction term*:

$$e_{ij} \approx \text{interaction between row } i \text{ and column } j.$$

Large residuals suggest:

- Possible interaction,
- Outlying cells,
- Model misfit.

If the additive model is appropriate, residuals should be small and structureless.

## 7.4 Example: Infant Mortality (Region × Education)

Rows = U.S. region Columns = father's education level Entries = infant mortality rates

### 7.4.1 Cleveland Dot Plot (Quick Visualization)

```
df = df %>% as.matrix()
dotchart(df)
```

The dot plot visually reveals:

- Highest mortality in Southern region.
- Lowest mortality in highest education category.
- Possible non-parallel patterns suggesting interaction.

### 7.4.2 Median Polish Output Interpretation

After running median polish:

- $\theta$ represents baseline infant mortality.
- Row effects quantify regional differences.
- Column effects quantify education effects.
- Large residuals highlight unusual region–education combinations.

For example: A large positive residual for "Southern region × low education" suggests mortality higher than predicted by additivity alone.

## 7.5 Diagnostics for Additivity

Median polish assumes approximate additivity. Diagnostics help assess whether this is reasonable.

### 1. Tukey Additivity Plot

A scatter plot of residuals versus fitted additive values:

$$\hat{X}_{ij} = \theta + \beta_i + \tau_j.$$

A visible systematic trend suggests:

- Interaction between row and column factors,
- Need for transformation (e.g., log scale),
- Possible multiplicative structure.

**2. Profile Plots**

Plot row profiles across columns (or vice versa).

- Parallel lines $\Rightarrow$ additive structure.
- Crossing lines $\Rightarrow$ interaction.

If strong interaction is present, one may consider:

- transformation of the response (e.g., $\log X_{ij}$),
- inclusion of interaction terms,
- alternative nonparametric smoothing approaches.

### 7.6 Summary

- Median polish is a robust alternative to two-way ANOVA.
- Uses medians instead of means.
- Iteratively removes row and column medians.
- Residuals diagnose interaction and outliers.
- Primarily exploratory, not inferential.

From a broader perspective:

- Median polish is a robust additive decomposition method,
- Closely related to $L_1$-type estimation,
- Useful as a preprocessing or exploratory step before formal modeling.

## 8 PERMANOVA: Permutational Multivariate ANOVA

### 8.1 Motivation

In many modern applications, the response is not a scalar but a *multivariate object*. Examples include:

- microbial community compositions (relative abundances of taxa),
- gene expression profiles (thousands of genes),
- brain connectivity matrices (network weights),
- high-dimensional feature vectors from imaging or omics data.

The scientific question is often:

*Do two or more groups differ in their overall multivariate structure?*

For instance:

- Do microbiomes differ between healthy and diseased subjects?
- Do brain connectomes differ across diagnostic groups?
- Do treatment groups differ in multivariate biomarker profiles?

Classical MANOVA addresses such questions but requires:

- multivariate normality,
- homogeneous covariance matrices,
- sample size larger than dimension.

These assumptions are often unrealistic in modern high-dimensional data. PERMANOVA provides a flexible alternative.

## 8.2 What is PERMANOVA?

**PERMANOVA** stands for:

**Permutational Multivariate Analysis of Variance**.

It is a *distance-based*, nonparametric method that tests whether group centroids differ in a multivariate space.

Rather than modeling raw vectors directly, PERMANOVA works with a distance (or dissimilarity) matrix:

$$D = (d_{ij}), \quad d_{ij} = \text{distance between subjects } i \text{ and } j.$$

Common choices of distance include:

- Euclidean distance (continuous features),
- Bray–Curtis dissimilarity (ecology),
- UniFrac distance (microbiome phylogenetic data),
- correlation-based distances (connectivity matrices).

Thus, PERMANOVA is extremely flexible: any meaningful dissimilarity measure may be used.

Importantly, the method depends only on pairwise distances, and therefore does not require explicit embedding of the data in Euclidean space.

### 8.3 Mathematical Connection to ANOVA

Recall classical ANOVA:

$$\text{Total SS} = \text{Between-group SS} + \text{Within-group SS}.$$

PERMANOVA generalizes this idea using distances.

Let $n$ denote the total number of observations, partitioned into $G$ groups.

Define the total sum of squares in terms of distances:

$$\text{SS}_T = \frac{1}{n} \sum_{i<j} d_{ij}^2.$$

**Gower centering.** Define the matrix

$$\mathbf{A} = -\frac{1}{2} D^{\circ 2},$$

where $D^{\circ 2}$ denotes elementwise squared distances.

Let

$$\mathbf{H} = I_n - \frac{1}{n} \mathbf{1}\mathbf{1}^T$$

be the centering matrix.

Then the centered matrix

$$\mathbf{G} = \mathbf{HAH}$$

is a Gram matrix (inner product matrix) when distances are Euclidean.

Thus, PERMANOVA operates on $\mathbf{G}$, which plays the role of a covariance matrix in multivariate analysis.

### 8.4 Projection Formulation

Let $\mathbf{X}$ be the design matrix encoding group membership, and let $\mathbf{P} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ be the projection matrix onto the group space.

Then:

$$\text{SS}_B = \text{tr}(\mathbf{PG}), \qquad \text{SS}_W = \text{tr}((I - \mathbf{P})\mathbf{G}).$$

Thus,

$$\text{SS}_T = \text{tr}(\mathbf{G}) = \text{SS}_B + \text{SS}_W.$$

This is directly analogous to linear-model ANOVA, but applied to the distance-induced inner product matrix.

## 8.5 Test Statistic

The PERMANOVA test statistic is an $F$-type ratio:

$$F = \frac{\mathrm{SS}_B/(G-1)}{\mathrm{SS}_W/(n-G)}.$$

Equivalently, in matrix form:

$$F = \frac{\mathrm{tr}(\mathbf{PG})/(G-1)}{\mathrm{tr}((I-\mathbf{P})\mathbf{G})/(n-G)}.$$

Large values of $F$ indicate that between-group dissimilarities dominate within-group dissimilarities.

Importantly:

- No distributional assumptions are made about the data.
- The statistic depends entirely on the chosen distance.

## 8.6 Permutation-Based Inference

Under the null hypothesis:

$$H_0: \text{ group labels do not affect the multivariate distribution},$$

group labels are exchangeable.

More formally:

$$\{X_i\}_{i=1}^n \text{ are i.i.d. up to group labels.}$$

PERMANOVA proceeds as follows:

1. Compute observed statistic $F_{\mathrm{obs}}$.
2. Randomly permute group labels $B$ times.
3. For each permutation $b$, compute $F^{(b)}$.
4. Estimate p-value:
$$p = \frac{1 + \#\{F^{(b)} \geq F_{\mathrm{obs}}\}}{1+B}.$$

This produces an exact (or Monte Carlo) permutation test.

Because inference is based on label permutations, PERMANOVA is fully nonparametric and valid under exchangeability.

### 8.7 Geometric Interpretation

PERMANOVA can be viewed geometrically:

- Each subject lies in a high-dimensional space (possibly implicit).
- $\mathbf{G}$ encodes inner products between centered observations.
- Group centroids correspond to projected means in this space.
- The test measures separation between projected group centroids.

When Euclidean distance is used, PERMANOVA becomes equivalent to classical MANOVA under certain conditions.

More generally, this framework connects to kernel methods: $\mathbf{G}$ may be viewed as a kernel matrix, placing PERMANOVA within an RKHS-based testing framework.

### 8.8 Important Caveat: Dispersion vs Location

A crucial limitation:

> PERMANOVA detects differences in multivariate spread, not only differences in centroids.

Mathematically,

$$\mathrm{SS}_B = \mathrm{tr}(\mathbf{PG})$$

can increase due to:

- differences in group means (location),
- differences in within-group covariance (dispersion).

Thus, a significant result may arise because:

- Group centroids differ (true location effect), or
- One group has larger within-group dispersion.

To assess dispersion effects, researchers often perform:

$$\text{PERMDISP} \quad \text{(permutation test for homogeneity of dispersions).}$$

If dispersions differ substantially, PERMANOVA results may reflect variance differences rather than mean shifts.

### 8.9 Extensions

PERMANOVA generalizes easily to:

- Multiple factors (two-way layouts),
- Interaction effects,
- Covariate adjustment,
- Nested designs.

It resembles linear model ANOVA, but all computations occur in distance space.

## 8.10 Software

In `R`, PERMANOVA is implemented via:

- `adonis()` or `adonis2()` in the `vegan` package.

  Example:

```
############################################################
# Load vegan and data
############################################################
library(vegan)

data(dune)       # species abundance (community data)
data(dune.env)   # environmental metadata

############################################################
# Create distance matrix
############################################################
D <- vegdist(dune, method = "bray")

############################################################
# Define grouping variable
############################################################
group <- dune.env$Management   # factor with groups

############################################################
# PERMANOVA
############################################################
adonis2(D ~ group, data = dune.env, permutations = 999)
```

  For dispersion testing:

```
bd <- betadisper(D, group)
bd

permutest(bd)
```

## 8.11 Summary

- PERMANOVA is a distance-based multivariate test.
- It generalizes ANOVA to arbitrary dissimilarities.
- Inference relies on permutations.
- It is widely used in microbiome, ecology, imaging, and genomics.
- Significant results may reflect location or dispersion differences.
- Always consider complementary dispersion diagnostics.

## 9 Summary: Which Method When?

The choice of method depends on:

- number of groups,
- independence vs. blocking structure,
- completeness of the design,
- whether a structured alternative is present,
- whether the goal is inference or exploration.

More fundamentally, all methods in this chapter follow a common principle:

> *Define a notion of variation (via data or distances), decompose it into systematic and residual components, and assess whether the systematic component is larger than expected under a null model of exchangeability.*

Below is a practical decision guide.

- **One-way layout, $k \geq 3$ independent groups, general differences:**
  Use **Kruskal–Wallis** (KW) test (1).
  - Nonparametric analogue of one-way ANOVA.
  - Tests equality of group locations.
  - Assumes independent samples and continuous distributions.
  - Invariance under monotone transformations.
  - Large-sample reference: $\chi^2_{k-1}$.

  **When it works best:**
  - Differences are primarily in location (median shifts).
  - Groups have similar shapes and dispersions.

  **Potential limitation:**
  - Sensitive to differences in spread, not purely location.

- **One-way with structured alternatives:**
  If prior scientific knowledge suggests structure, use specialized tests:
  - **Ordered (monotone) alternative:** Jonckheere–Terpstra test. More powerful when a trend is expected.
  - **Umbrella alternative (increase then decrease):** Mack–Wolfe test.
  - **Treatments vs. control:** Fligner–Wolfe or similar control-focused procedures.

**Key idea:** Power is increased by restricting the alternative space.

**Trade-off:** Misspecification of the structure can reduce power.

- **After global rejection (one-way):**
  Identify which groups differ:

  - Perform pairwise Wilcoxon rank-sum (Mann–Whitney) tests.
  - Adjust $p$-values to control familywise error rate (FWER).
  - Use Bonferroni (simple, conservative) or Holm (more powerful).

  This answers: *Where are the differences?*

  **Interpretation:** Each pairwise test estimates

  $$\Pr(X_i < X_j),$$

  providing a probabilistic interpretation of group differences.

- **Two-way randomized complete block design (RCBD):**
  Use **Friedman test** (2).

  - Nonparametric analogue of two-way ANOVA without interaction.
  - Blocks remove nuisance variability.
  - Rank within each block (eliminates block effects).
  - Large-sample reference: $\chi^2_{k-1}$.
  - Apply tie correction if necessary.

  **When it works best:**

  - Strong block effects (heterogeneity across subjects/units).
  - Treatments compared within homogeneous blocks.

  **Limitation:**

  - Cannot handle missing treatments within blocks.
  - Does not model interaction explicitly.

  Follow-up:

  - Paired comparisons (signed-rank or sign test).
  - Adjust for multiplicity (Holm recommended).

- **Incomplete blocks / missing cells:**
  Use **Skillings–Mack** test.

  - Generalizes Friedman.

- Allows unequal treatment representation across blocks.
- Uses normalized, centered ranks within each block.
- Asymptotic $\chi^2_{k-1}$ reference.

**Key advantage:**

- Maintains validity under unbalanced or partially observed designs.

**Interpretation:**

- Same as Friedman, but with weighting to account for missingness.

- **Exploratory robust decomposition of two-way tables:**
  Use **Median polish**.

  - Robust additive decomposition via medians.
  - Separates overall, row, and column effects.
  - Highlights unusual residual cells.
  - Primarily exploratory, not inferential.

  Diagnostic tools:

  - Tukey additivity plot.
  - Profile plots.
  - Dot plots / boxplots for visualization.

  **Key interpretation:**

  - Residuals represent interaction or model misfit.
  - Large residuals indicate departures from additivity.

- **Multivariate responses (distance-based comparisons):**
  Use **PERMANOVA**.

  - Distance-based multivariate analogue of ANOVA.
  - Operates on dissimilarity matrices.
  - Permutation-based inference.
  - Equivalent to quadratic form in a Gram (kernel) matrix.

  **When it works best:**

  - High-dimensional or non-Euclidean data.
  - Meaningful distance metric is available.

  **Caution:**

  - Sensitive to both location and dispersion differences.
  - Always check dispersion using PERMDISP.

**Big Picture**

- Independent groups $\Rightarrow$ Kruskal–Wallis.
- Blocking $\Rightarrow$ Friedman.
- Incomplete blocking $\Rightarrow$ Skillings–Mack.
- Structured alternative $\Rightarrow$ specialized rank test.
- Post hoc comparisons $\Rightarrow$ Wilcoxon + FWER control.
- Exploratory two-way structure $\Rightarrow$ Median polish.
- Multivariate distance data $\Rightarrow$ PERMANOVA.

**Unifying perspective.**
All methods in this chapter can be written in the form:

$$\text{Statistic} = \text{quadratic form measuring systematic variation} \Big/ \text{residual variation.}$$

More concretely:

- ANOVA: quadratic forms in raw data.
- KW / Friedman / Skillings–Mack: quadratic forms in ranks.
- PERMANOVA: quadratic forms in distance-induced inner products.

All methods preserve the core ANOVA logic:

$$\text{Compare systematic variation to residual variation,}$$

while relaxing parametric distributional assumptions.

**Final takeaway:**

> Choose the method based on the *design structure* first, then the *type of response* (scalar vs. multivariate), and finally the *scientific question* (global vs. structured vs. exploratory).

# References

[Chatfield and Mander(2009)] Mark Chatfield and Adrian Mander. The skillings–mack test (friedman test when there are missing data). *The Stata Journal*, 9(2):299–305, 2009.